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Bachelor Thesis

Evaluating and Optimizing Text-Embedding Quality for German Tax Law Retrieval

Data Science and Artificial Intelligence

Author:	Felix Niklas Regler
Student ID:	9445603
Course:	WDSKI23B
Company:	PricewaterhouseCoopers GmbH WPG
Department:	Tax & Legal Digital
Department-Supervisor:	Sebastian Ziesing (Manager, PwC)
Product Owner:	Prof. Dr. Arne Schnitger (Partner, PwC)
Academic Director:	Prof. Dr. Drabant
Academic Supervisor:	Prof. Dr. Scherer
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Declaration of Authorship

I hereby declare that I have written the present work entitled “*Evaluating and Optimizing Text-Embedding Quality for German Tax Law Retrieval*” independently and have not used any sources or aids other than those indicated.

Place, Date

Felix Niklas Regler

Abstract

Author: Felix Niklas Regler
Student ID: 9445603
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Retrieval-Augmented Generation systems in highly regulated domains such as German tax law depend on the quality of their embedding-based retrieval stage, yet standard commercial embedding models suffer from a well-documented domain gap when applied to specialised legal text. The lexical density, morphological complexity and fine-grained semantic distinctions of German tax law induce geometric pathologies in general-purpose embedding spaces ...

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List of Abbreviations

ABTT	All-but-the-Top
AI	Artificial Intelligence
ANN	Approximate Nearest Neighbour
BERT	Bidirectional Encoder Representations from Transformers
BFH	Bundesfinanzhof
BMF	Federal Ministry of Finance
BoW	Bag of Words
CoT	Chain-of-Thought
CV	Coefficient of Variation
DCG	Discounted Cumulative Gain
DRM	Document-Level Retrieval Mismatch
DSR	Design Science Research
DSRM	Design Science Research Methodology
GenAI	Generative Artificial Intelligence
GerDaLIR	German Dataset for Legal Information Retrieval
GloVe	Global Vectors for Word Representation
HNSW	Hierarchical Navigable Small World
IQR	Interquartile Range
IR	Information Retrieval
LIR	Legal Information Retrieval
LLM	Large Language Model
LOO	Leave-One-Out
LSP	Language for Specific Purposes
MIPS	Maximum Inner Product Search
MLM	Masked Language Modelling
MRR	Mean Reciprocal Rank
nDCG	Normalized Discounted Cumulative Gain
NEO	Tax Analysis Generator
NLP	Natural Language Processing
OOV	Out-of-Vocabulary
PC	Principal Component
PCA	Principal Component Analysis

PR	Participation Ratio
PwC	PricewaterhouseCoopers GmbH WPG
RAG	Retrieval-Augmented Generation
RBO	Rank-Biased Overlap
SVD	Singular Value Decomposition
TF-IDF	Term Frequency-Inverse Document Frequency
TSDAE	Transformer-based Sequential Denoising Auto-Encoder
TwoNN	Two Nearest Neighbors
ZCA	Zero-phase Component Analysis

List of Notations

The following notation is used consistently throughout this thesis.

Symbol	Description
Corpora and Vocabularies	
$\mathcal{X}$	Input space of discrete linguistic objects
$\mathcal{C}$	A text corpus ($\mathcal{C} \subset \mathcal{X}$)
$\mathcal{C}_{\text{neo}}$	The NEO tax law corpus
$\mathcal{C}_{\text{ref}}$	A general-domain reference corpus e.g., Wikipedia
N	Total number of chunks in a corpus
$\mathcal{V}$	Vocabulary
$\mathcal{V}_{\mathcal{C}}$	Vocabulary of corpus $\mathcal{C}$
w	A word or sub-word token ($w \in \mathcal{V}$)
$f_{\mathcal{C}}(w)$	Frequency of token w in corpus $\mathcal{C}$
Linguistic and Morphological Properties	
$L(w)$	Character length of token w
$\bar{L}(\mathcal{V})$	Mean content-word length over vocabulary $\mathcal{V}$
τ	Tokeniser function ($\tau: \mathcal{X} \rightarrow \mathcal{V}^*$)
$F(w)$	Token fertility: sub-word tokens produced for w ($F(w) = \tau(w) $)
$\bar{F}(\mathcal{V})$	Mean token fertility over vocabulary $\mathcal{V}$
Embeddings and Retrieval Architecture	
<i>Representation space</i>	
$\mathcal{Z}$	Representation space ($\mathcal{Z} \subseteq \mathbb{R}^d$)
d	Ambient embedding dimensionality (e.g., $d = 3,072$)
d'	Reduced dimensionality after projection (e.g., $d' = 512$)
$\mathbf{z}$	Embedding vector of a text ($\mathbf{z} \in \mathbb{R}^d$)
$\boldsymbol{\mu}$	Corpus centroid (mean embedding vector)
$\boldsymbol{\Sigma}$	Covariance matrix of a set of embeddings
λ_i	i -th eigenvalue of $\boldsymbol{\Sigma}$
<i>Encoder components</i>	
$\phi_{\theta}(\cdot)$	Parameterised embedding model
$\Phi(\cdot)$	Full encoder pipeline ($\Phi: \mathcal{X} \rightarrow \mathcal{Z}$)
$\boldsymbol{\Omega}$	Global embedding matrix for static paradigms ($\boldsymbol{\Omega} \in \mathbb{R}^{d \times \mathcal{V} }$)
$\mathbf{E}$	Token embedding matrix ($\mathbf{E} \in \mathbb{R}^{d \times n}$)
$\psi(\cdot)$	Pooling function, mapping token embeddings to a sequence vector
$\mathbf{W}$	PCA projection matrix ($\mathbf{W} \in \mathbb{R}^{d \times d'}$)
<i>Retrieval</i>	
q	A user query
c	A text chunk from $\mathcal{C}$
k	Retrieval depth
$\mathcal{Q}$	Evaluation query set
$\mathcal{R}_q$	Ground-truth relevant documents for query q
$\mathcal{C}_{q,k}$	Top- k retrieved chunks for query q
$L_{q,k}$	Ranked retrieval list: $L_{q,k} = (c^{(1)}, \dots, c^{(k)})$

Symbol	Description
rel_i	Graded relevance score at rank i in $L_{q,k}$
$\hat{\mathbf{Q}}$	Query embedding matrix ($\hat{\mathbf{Q}} \in \mathbb{R}^{ \mathcal{Q} \times d'}$)
$\hat{\mathbf{C}}$	Corpus embedding matrix ($\hat{\mathbf{C}} \in \mathbb{R}^{N \times d'}$)
$\text{sim}(\mathbf{u}, \mathbf{v})$	Cosine similarity between $\mathbf{u}$ and $\mathbf{v}$
Geometric Diagnostics and Evaluation Metrics	
<i>Intrinsic (geometric) metrics</i>	
$\bar{s}_{\text{cos}}$	Mean pairwise cosine similarity (angular concentration)
D_μ	Mean dominance ratio (centroid bias)
IsoScore	Eigenvalue uniformity of Σ (spatial isotropy)
I_{PF}	Partition function isotropy (directional uniformity)
$\hat{A}_{\text{centered}}$	Centered anisotropy (post-mean-subtraction variance distribution)
PR	Participation ratio (effective intrinsic dimensionality)
R_λ	Eigenvalue ratio λ_1/λ_d (spectral condition number)
$\hat{d}_{\text{MLE}}$	MLE estimator of local intrinsic dimensionality
$\mathcal{L}_{\text{uniform}}$	Uniformity loss on the unit hypersphere
S_k	Hubness skewness of the k -NN occurrence distribution
Δ_{sep}	Inter-corpus separation gap
<i>Extrinsic (retrieval) metrics</i>	
nDCG@ k	Normalised Discounted Cumulative Gain at rank k
MRR	Mean Reciprocal Rank
P@ k	Precision at rank k
Statistical and Experimental Notation	
T	Number of independent repetitions
B	Number of bootstrap resamples
α_{adj}	Bonferroni-adjusted significance level (α/m)
$y_{i,j}$	Graded relevance label for query q_i , chunk c_j
g	Hedge's g (bias-corrected effect size)
κ	Cohen's kappa (inter-rater agreement)
ρ	Spearman's rank correlation coefficient

1 Introduction

The emergence of Generative Artificial Intelligence (GenAI) and its most prominent representatives, Large Language Models (LLMs), marked a technological inflection point. Over the past few years, this self-accelerating development has catalysed a profound transformation across a wide range of industries, changing the way knowledge is aggregated, processed and applied [1]. Even historically conservative and heavily regulated sectors such as accounting and tax consultancy are no exception. Law and consultancy firms face pressure on margins and fees, compelling them to adopt these technologies not as incremental aids but as strategic imperatives for operational efficiency [2]. At the centre of this development lies the ambition to provide automated, intelligent support for time-consuming cognitive routine tasks, such as drafting tax memoranda or conducting initial legal assessments of complex cases [3–5].

Yet the regulatory and technical demands of tax consultancy impose constraints that fundamentally distinguish it from the creative text-generation tasks at which LLMs excel. Legal and tax advice requires absolute factual accuracy and normative precision [6]. An erroneous statement can trigger extensive financial liability and regulatory consequences for both the adviser and the client [7]. However, standard LLMs are vulnerable to hallucination¹ and cannot reliably apply highly specialised or up-to-date expert knowledge, such as a firm’s proprietary interpretations of current legislation [1, 8]. Since continuous fine-tuning of the models to update knowledge is too inefficient and inflexible in industrial practice, alternative system architectures are required [1, 9–11].

Against this background and in the absence of any market-available Artificial Intelligence (AI) solution that simultaneously satisfies the stringent requirements for legally valid advice and adaptively integrates proprietary in-house knowledge, PricewaterhouseCoopers GmbH WPG (PwC) developed the Tax Analysis Generator (NEO). NEO is based on the paradigm of Retrieval-Augmented Generation (RAG), a design that overcomes the limitations of a static parametric memory by externalising factual knowledge into an external corpus. When a user makes a request, an Information Retrieval (IR) step first identifies the most relevant passages from the underlying corpus and injects them as grounding context into the LLM’s generation process, thereby anchoring its output in source material [11].

While the discourse about LLM hallucinations dominates the generative AI landscape [8], this thesis shifts attention to the critical upstream bottleneck, the retrieval process itself. The quality of a RAG system such as NEO is bounded by the quality of its document search, if the retrieval algorithm fails to find the correct information, even the best generator will inevitably produce inadequate or misleading output [10, 12, 13]. Currently, NEO uses a standard commercial embedding model (OpenAI’s `text-embedding-3-large`) for dense retrieval. This universal embedder is widely adopted across the industry and trained to process a broad spectrum of texts, from poetry to recipes to news articles, a generality that is simultaneously its strength and its potential liability [14].

Nevertheless, the German tax law domain poses fundamentally different requirements. Tax law texts are characterised by lexical homogeneity, formulaic phrasing and fine-grained semantic nuances, subtle distinctions in meaning that carry significant legal consequences [15–17]. It is a well-established finding in Natural Language Processing (NLP) that universal embedding models suffer from a domain gap in such specialised settings [18–20]. Their representation spaces undergo a form of geometric collapse in which, although the model can separate tax

¹The confident generation of plausible but factually incorrect content.

texts from general-language documents, it fails to resolve the intra-domain distinctions that determine retrieval relevance [21–23].

This problem is reinforced by the fact that in specialised domains such as German tax law, annotated reference datasets, the ground truth against which retrieval quality would conventionally be measured, are generally not available and their creation by domain experts is prohibitively expensive and time-consuming [5, 13, 24]. In practice, this makes it considerably more difficult to objectively evaluate whether a universal embedding model is at all suitable for a given use case and how it might be systematically optimised or replaced [25, 26].

The present work addresses this gap. Rather than relying exclusively on costly, manually labelled evaluation datasets, this thesis approaches the problem from the perspective of vector space geometry. The aim is twofold. Firstly, it seeks to investigate the extent to which established, universal embedding models are suitable for RAG-based retrieval in German tax law. Secondly, it examines whether the analysis of intrinsic geometric properties of embedding spaces can serve as a reliable predictor of actual, extrinsic retrieval performance. Appropriate quantitative metrics are applied to assess whether targeted interventions, ranging from domain-specific fine-tuning to resource-efficient post-processing transformations, can improve search quality measurably. From this objective, the central research question of this thesis emerges:

How can the suitability of embedding models for the German tax law domain be systematically evaluated and optimised?

To address this question in a structured manner, the investigation is decomposed into three guiding sub-questions:

1. Geometric Characterisation and Response: What geometric properties do standard embeddings exhibit in the German tax law domain and how do these properties respond to different optimisation strategies?
2. Predictive Validity: To what extent do intrinsic geometric properties correlate with actual retrieval quality and can they therefore serve as a reliable diagnostic signal in the absence of gold-standard evaluation data?
3. Retrieval Improvement: Which of the tested optimisation strategies, such as domain-specific fine-tuning, model substitution, or post-hoc geometric transformations, yields the greatest measurable improvement in task-related retrieval quality for this domain?

To delineate the scope of this study with precision, the work deliberately focuses exclusively on the dense retrieval step of the RAG pipeline. While overall system performance could be enhanced through various complementary pipeline optimisations, such as hybrid search strategies, query expansion or neural re-ranking [27], these extensions are intentionally excluded to avoid introducing confounding variables that would obscure the intrinsic capabilities of the embedding models under investigation. By holding the surrounding pipeline architecture constant, this thesis aims to derive fundamental, transferable recommendations for the selection, efficient optimisation and performance-investigation of embedding models in specialised industrial domains.

2 Theoretical Foundations

2.1 Text Representation and Embeddings

The primary objective in the field of NLP is to overcome the disparity between the unstructured nature of human language and the structured, numerical representations necessary for the functionality of computational systems [28, 29]. The way humans reason about language is through words, sentences and documents. However, these forms of language cannot be processed by machines, since they mostly operate on continuous vector spaces. Therefore, any computational process requires mapping the space of linguistic objects $\mathcal{X}$ into a mathematical target space $\mathcal{Z} \subseteq \mathbb{R}^d$ [28, 30][31, p. 374f.].

$$\Phi : \mathcal{X} \rightarrow \mathcal{Z} \quad (2.1)$$

This mapping Φ , visualized in Figure 2.1, is called an encoder and the resulting output $\mathbf{z} = \Phi(x)$ is the representation¹ of x . The goal is to construct Φ such that $\mathcal{Z}$ becomes a structured space where geometric relationships, such as distance or angle, reflect the semantic relationships of the original data in $\mathcal{X}$ [31, p. 374 f.][32, 33].

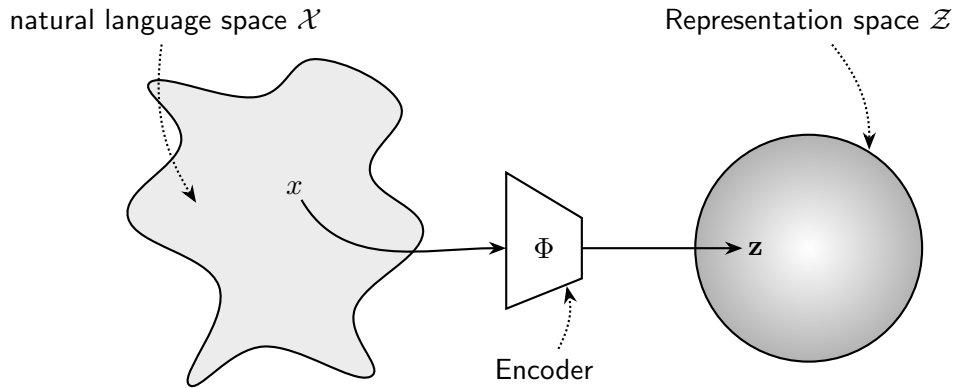


Figure 2.1: An encoder Φ maps discrete linguistic inputs x from the text space into continuous vectors $\mathbf{z}$ in a structured representation space.
(Own illustration based on Bengio et al. [29])

Modern NLP systems decompose this mapping into a distinct pipeline, formalized by

$$\underbrace{x}_{\text{raw text}} \xrightarrow{\rho} \underbrace{x'}_{\text{cleaned}} \xrightarrow{\tau} \underbrace{(w_1, \dots, w_n)}_{\text{tokens}} \xrightarrow{\phi_\theta} \underbrace{\mathbf{E} \in \mathbb{R}^{d \times n}}_{\text{token embeddings}} \xrightarrow{\psi} \underbrace{\mathbf{z} \in \mathbb{R}^d}_{\text{sequence embedding}}. \quad (2.2)$$

First, the preprocessing function ρ standardises raw text x from the input space $\mathcal{X}$ into a normalised format x' . Typically, this stage involves tasks such as lowercasing, Unicode normalisation and the removal of special characters or stopwords to reduce the variation of surface forms² [34, p. 4 f., p. 243]. If no cleaning is required, ρ simplifies to the identity function

¹In the literature, the terms *representation* and *embedding* are often used interchangeably. In this work, *representation* is used as the broader term, including deterministic constructions, while *embedding* is used specifically for learned vector spaces where ϕ_θ is optimised from data.

²Although this normalisation is crucial, as all subsequent stages rely on its consistency, it will not be discussed in depth in this work. These standard procedures are already extensively documented in foundational NLP literature [34, p. 13 f.]

$x' = x$. A specific, finite collection of texts is referred to as a corpus $\mathcal{C} \subset \mathcal{X}$. Second, a tokenizer τ segments x' into a sequence of discrete tokens based on a predefined vocabulary $\mathcal{V}$. Third, an embedding model ϕ_θ maps this discrete sequence into a continuous vector space, yielding a matrix of token-level embeddings $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_N]$. Finally, if downstream tasks require a fixed-size vector, an optional aggregation function ψ compresses these token-level representations into a single, global sequence embedding $\mathbf{z}$ [35]. Consequently, the overarching encoder principle defined in Equation 2.1 can be formally understood as the composition of these pipeline stages: $\Phi(x) = (\psi \circ \phi_\theta \circ \tau \circ \rho)(x)$ [34, 36].

2.1.1 Tokenization

A tokenizer $\tau : \mathcal{X} \rightarrow \mathcal{V}^*$ segments a raw text into a finite sequence over a predefined vocabulary $\mathcal{V}$, where $*$ denotes the Kleene Closure [34, p. 4 f.]. Hereby, the choice of $\mathcal{V}$ defines the granularity of the atomic units on which all subsequent stages operate:

- **Character-level** tokenization sets $\mathcal{V}$ to the character alphabet, eliminating Out-of-Vocabulary (OOV) items at the cost of substantially longer sequences and reduced ability to capture word-level semantics [37].
- **Word-level** tokenization segments text at whitespace or punctuation boundaries, producing linguistically intuitive units. It requires a very large vocabulary for adequate coverage and inevitably fails on morphological variants and neologisms absent from the fixed vocabulary. Moreover, it implicitly assumes that word boundaries are marked by whitespace, an assumption that does not hold for languages such as Chinese, Japanese, or Thai, where dedicated segmentation algorithms are required [34, p. 6 f.][38].
- **Sub-word** tokenization strikes a balance between both extremes and has become the *de facto* standard in modern NLP [37]. Rather than committing to a fixed word- or character-level vocabulary, sub-word methods learn variable-length units from a training corpus through iterative statistical procedures. Widely adopted algorithms include BPE [38], WordPiece [39] and the Unigram language model [40].
- **N-gram** tokenization decomposes text into overlapping sequences of n characters or words. While less common as a primary strategy for neural models, character n -grams serve as the representational basis for FastText [41], enabling meaningful representations even for OOV tokens.

A critical consequence of the tokenizer’s design is its interaction with the morphological properties of the target language. Sub-word tokenizers trained on general-domain corpora learn merge rules reflecting the frequency distribution of that data [38, 42]. When encountering specialised-domain text whose vocabulary diverges from the training distribution, domain-specific terms are fragmented into disproportionately many sub-word units. This effect is captured by token fertility $F(w) = |\tau(w)|$, the number of sub-word tokens a word w is decomposed into [37]. High fertility is particularly problematic for morphologically rich languages such as German, where productive compounding generates forms absent from general-purpose vocabularies [42].

2.1.2 Embedding Paradigms

Frequency-Based Embeddings. The most straightforward approach to representing text numerically is to assign each token $w \in \mathcal{V}$ a unique identifier, such as a random number or an index in the vocabulary, which can be represented as a one-hot vector of the length of the Vocabulary $\mathbf{v}_w \in \{0, 1\}^{|\mathcal{V}|}$. Because the order of the vocabulary or the chosen numbering is arbitrary, there is no semantic structure in this space. Further, since every token is assigned to its own orthogonal axis, the distance between any two distinct token vectors carries no meaningful information [31, p. 374 f.]. In fact, the dot product between any two distinct token

vectors is always zero, thereby precluding the capture of any semantic similarity or relationships between different words [43, p. 22 f.].

Consequently, traditional statistical methods mostly bypass token-level representations entirely and directly aggregate the sparse token vectors into a single, global document representation $\mathbf{z}$. The assumption of these methods is that texts $x_i \in \mathcal{X}$ can be handled as an unordered collection of tokens, disregarding syntax and word order [31, p. 378][44, p. 11]. In the foundational Bag of Words (BoW) model, each dimension of the resulting vector corresponds simply to the raw Term Frequency $\text{TF}(w, x_i) = f_{w, x_i}$ of a specific token within the given text $\mathbf{z} = (\text{TF}(w_1, x_i), \dots, \text{TF}(w_{|\mathcal{V}|}, x_i))$. Nevertheless, these raw frequencies disproportionately emphasize common but semantically uninformative tokens, such as articles or prepositions. To mitigate this, the Term Frequency-Inverse Document Frequency (TF-IDF) model adjusts the local term frequency by incorporating the inverse document frequency $\text{IDF}(w, \mathcal{C}) = \log \left(\frac{|\mathcal{C}|}{|\{d \in \mathcal{C} : w \in d\}|} \right)$, which measures how rare and discriminative a given token is across an entire corpus, resulting in $\mathbf{z} = (\text{TF}(w_1, x_i) \times \text{IDF}(w_1, \mathcal{C}), \dots, \text{TF}(w_{|\mathcal{V}|}, x_i) \times \text{IDF}(w_{|\mathcal{V}|}, \mathcal{C}))$ [34, p. 101 f.][36, p. 117 f.][43, p. 23 f., p. 69][44, 45].

Static Semantic Embeddings. While these statistical methods represent text as highly sparse, high-dimensional vectors, their fundamental inability to capture the underlying semantic relationships between different words motivates a transition to dense representations. Semantic embedding methods overcome this limitation by mapping tokens into a dense continuous vector space $\mathbb{R}^d$ where $d \ll |\mathcal{V}|$, such that geometric proximity reflects semantic similarity [32, 43]. Rooted in the distributional hypothesis, which states that words occurring in similar contexts tend to have similar meanings, these methods assign a fixed representation to every element in the vocabulary $\mathcal{V}$ [34, p. 96 f.].

At its core, methods following this paradigm construct a global embedding matrix $\mathbf{\Omega} \in \mathbb{R}^{d \times |\mathcal{V}|}$, where each column $\boldsymbol{\omega}_w \in \mathbb{R}^d$ represents the continuous vector for the w -th vocabulary item, or token. In order to retrieve the embedding for a specific token w , the token is first represented as a one-hot encoded vector $\mathbf{v}_w \in \{0, 1\}^{|\mathcal{V}|}$ and subsequently multiplied with the embedding matrix to extract the corresponding column as the dense embedding $\mathbf{e}_w = \mathbf{\Omega} \mathbf{v}_w = \boldsymbol{\omega}_w$. It follows that an entire sequence of n tokens can be generated by the matrix multiplication of the aggregated one-hot vectors $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_n] \in \{0, 1\}^{|\mathcal{V}| \times n}$ with the embedding matrix, yielding the embedding for the entire sequence³ $\mathbf{E} = \mathbf{\Omega} \mathbf{V} = [\mathbf{e}_1, \dots, \mathbf{e}_n] \in \mathbb{R}^{d \times n}$ [31, p. 375][34, p. 133 f.][43, p. 89 f., p. 124 f.][46, p. 218]. Although these algorithms were historically developed and named for word representations, they operate abstractly on any atomic units defined by the tokenizer’s vocabulary $\mathcal{V}$. Two prominent algorithms established this static paradigm by learning $\mathbf{\Omega}$ through different approaches [31, p. 375].

First, Word2Vec by Mikolov et al. [32] learns the embedding matrix $\mathbf{\Omega}$ through a local, predictive task using a shallow neural network. In its most popular variant, the Skip-Gram architecture, the model iterates over a corpus of length n and attempts to maximize the probability of predicting the surrounding context tokens w_{j+i} given a central token w_j within a sliding window of size c which is formalized as

$$\max_{\mathbf{\Omega}} \prod_{j=1}^n \prod_{-c \leq i \leq c, i \neq 0} \mathbb{P}(w_{j+i} \mid w_j; \mathbf{\Omega}) \quad (2.3)$$

where the probability is typically defined via a softmax function over the inner products of the token vectors. Throughout the maximization process, the column vectors $\boldsymbol{\omega}_w$ of the matrix $\mathbf{\Omega}$

³This matrix-vector formulation is a conceptual device that clarifies the linear-algebraic structure of the embedding lookup. In practice, implementations avoid the multiplication with a sparse one-hot vector entirely and retrieve $\boldsymbol{\omega}_w$ directly via an index-based table lookup, which reduces the operation from $\mathcal{O}(d \times |\mathcal{V}|)$ to $\mathcal{O}(1)$ [34, p. 133 f.].

are iteratively adjusted, which results in tokens that frequently share the same neighbourhood being pulled closer together in the representation space [32][34, p. 108 f.][43, p. 124 f.][47].

In contrast to this approach, the other popular method, Global Vectors for Word Representation (GloVe) by Pennington et al. [48], operates directly on global corpus statistics. It first constructs a global co-occurrence matrix $\mathbf{X}$, where X_{ij} denotes how often token w_i appears in the context of token w_j across the entire corpus. GloVe then frames the learning process as a matrix factorization problem, adjusting the embedding vectors such that their dot product perfectly approximates the logarithm of their probability of co-occurrence⁴ $\omega_i^\top \tilde{\omega}_j \approx \log X_{ij}$ [43, p. 127][48].

Both Word2Vec and GloVe successfully capture deep semantic relationships, famously allowing for algebraic operations in the vector space e.g. $\omega_{\text{king}} - \omega_{\text{man}} + \omega_{\text{woman}} \approx \omega_{\text{queen}}$ [32][34, p. 112 f.].

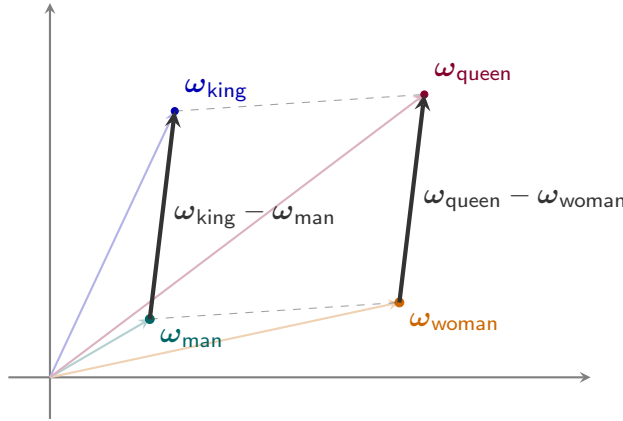


Figure 2.2: Geometric visualization of the algebraic word analogy. The parallel translation vectors illustrate that the geometric difference $\omega_{\text{king}} - \omega_{\text{man}}$ is approximately equal to $\omega_{\text{queen}} - \omega_{\text{woman}}$, which enables arithmetic operations within the semantic vector space. (Own illustration)

Contextual Semantic Embeddings. A major deficit of these static embedding methods is their static nature and context independence. Once training is complete, the matrix Ω is fixed and each token w_i is consistently mapped to the exact same vector ω_i regardless of its surroundings. The resulting semantic distortion is particularly obvious for homonyms and polysemes, since different word meanings will collapse into a single representation [21][34, p. 174 f.] eg. the word *book* will have the same embedding whether it refers to a physical object or the act of reserving something. Contextualized embedding models address this limitation by dynamically conditioning token representations on their surrounding context [34, p. 174 f.][49].

Formally, any sequence of n tokens $W = (w_1, \dots, w_n)$ is no longer processed through independent, fixed lookups. Instead, an embedding model maps the entire sequence jointly into a matrix of contextualised token representations $\phi_\theta(W) = [\mathbf{e}_1, \dots, \mathbf{e}_n] = \mathbf{E} \in \mathbb{R}^{d \times n}$, where each vector $\mathbf{e}_i$ depends not only on the identity of token w_i but on all other tokens in the sequence. Consequently, the same token receives different representations in different contexts, directly resolving the polysemy and homonymy problem that static methods cannot address [21, 49].

The foundational breakthrough enabling this dynamic representation was the introduction of the Transformer architecture by Vaswani et al. [50]. As a deep neural network, the Transformer is composed of multiple stacked layers that progressively refine the input representations, but

⁴For conceptual clarity, this equation isolates the core geometric relationship. The complete objective function optimised by GloVe additionally incorporates scalar bias terms for both the target and context tokens to account for their respective base frequencies within the corpus [48].

unlike its recurrent and convolutional predecessors, it entirely forgoes sequential processing in favour of a fully parallelisable mechanism known as self-attention [31, p. 357 f.]. Intuitively, self-attention is the process of determining, for each token in the sequence, which other tokens are relevant to it and how strongly they should influence its representation [31, p. 357 f.][43, p. 208 f.][50].

Because self-attention processes all tokens simultaneously as an unordered set, it inherently lacks any notion of sequential order. To inject positional information, the input tokens are first mapped to initial static semantic token embeddings $\tilde{\mathbf{E}}$ and enriched with positional encodings $\mathbf{P} \in \mathbb{R}^{d \times n}$, producing $\mathbf{E}_{\text{in}} = \tilde{\mathbf{E}} + \mathbf{P}$ as the input to the Transformer layers [34, p. 174 f.][46, p. 213 f.][50]. A single self-attention layer transforms $\mathbf{E}_{\text{in}}$ by applying three learned linear projections to produce the Query ($\mathbf{Q}$), Key ($\mathbf{K}$) and Value ($\mathbf{V}$) matrices:

$$\mathbf{Q} = \mathbf{W}_Q^\top \mathbf{E}_{\text{in}}, \quad \mathbf{K} = \mathbf{W}_K^\top \mathbf{E}_{\text{in}}, \quad \mathbf{V} = \mathbf{W}_V^\top \mathbf{E}_{\text{in}},$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d_k}$ are trainable weight matrices [46, p. 212 f.][50]. These projections serve different roles in the attention mechanism. The Query vector of a given token encodes what information this token is looking for, while the Key vectors characterise what each other token represents. The dot product between a Query and all Keys yields attention scores that quantify the relevance of each sequence position to the querying token. These scores are then used to compute a weighted sum over the Value vectors, which carry the actual content that each token contributes to the output [31, p. 362 f.][50]. The full operation is formalised as scaled dot-product attention

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}^\top \mathbf{K}}{\sqrt{d}}\right) \mathbf{V}.$$

The scaling factor $\sqrt{d}$ prevents the dot products from growing large enough to push the softmax into regions of vanishing gradients, while the softmax normalisation ensures that the resulting weights form a probability distribution over the sequence. Each token's output representation is thus a weighted combination of all Value vectors in the sequence, where contextually relevant tokens receive higher weights [31, p. 366 f.][46, p. 214][50].

To increase the representational capacity, multiple such attention operations are computed in parallel, a mechanism known as Multi-Head Attention. Allowing the model to attend to different types of relationships simultaneously e.g., syntactic dependencies in one head, semantic associations in another [51]. The outputs of all heads are concatenated and linearly projected back to the model dimension [31, p. 366 f.][34, p. 176 f.][50]. Each Transformer layer further comprises residual connections that preserve information from earlier layers, layer normalisation that stabilises training and a position-wise feedforward network that enables the learning of more complex, non-linear patterns. By stacking multiple such layers, the Transformer builds increasingly abstract, contextualised representations that capture both local syntactic patterns and long-range semantic dependencies [34, p. 191 f.][50][51].

The original Transformer was designed as an encoder-decoder architecture for sequence-to-sequence tasks such as machine translation [43, p. 208 f.][50]. The encoder transforms the input sequence into contextualised representations, while the decoder generates the output sequence token by token [43, p. 198 f.][50]. Further research has indicated that each component can be used independently, giving rise to two model families. Decoder-only models e.g., the GPT family by Radford et al. [52] employ causal (left-to-right) self-attention to generate text autoregressively [31, p. 383 f.]. In contrast, encoder-only models employ bidirectional self-attention across the full input sequence. This makes them particularly well suited for tasks that require rich contextual representations of existing text rather than the generation of new text [31, p. 388 f.][39].

The most prominent encoder-only model is Bidirectional Encoder Representations from Transformers (BERT) by Devlin et al. [39]. To learn its bidirectional representations, BERT is trained through two self-supervised objectives [46, p. 219 f.]. The primary task is Masked Language Modelling (MLM), where a random subset of input tokens (15%) is selected, of which 80% are replaced by a special [MASK] token, 10% by a random token and 10% are left unchanged. The model is then trained to reconstruct the original tokens based on the entire surrounding context, which is formalised as the maximisation of the conditional probability of the masked tokens $\mathcal{M} \subseteq \{1, \dots, n\}$ given an unmasked sequence $\max_{\theta} \prod_{i \in \mathcal{M}} \mathbb{P}_{\theta}(w_i | W_{\setminus \mathcal{M}})$ [31, p. 388][34, p. 202 f.][39]. The second training task, Next Sentence Prediction, trains the model to determine whether two given sentences are consecutive in the original text, thereby encouraging the learning of cross-sentence coherence [34, p. 204 f.][39]. Together, these objectives enable BERT to learn contextualised representations $\mathbf{E}_{\text{out}} = [\mathbf{e}_1, \dots, \mathbf{e}_n]$ that encode intra-sentence semantics as well as inter-sentence relationships [51].

Following BERT, many architectural variations and refinements have been proposed, including RoBERTa by Liu et al. [53], which optimises the training procedure on larger corpora and DeBERTa by He et al. [54], which detaches content and positional information in the attention mechanism. But all share the defining property that each token’s representation is a dynamic function of its entire sequence context [34, p. 207 f.].

2.1.3 Sequence Embeddings

For many applications, token-level representations are insufficient and a single, fixed-dimensional vector that captures the meaning of an entire sequence is required. To achieve this, an aggregation function ψ can be applied to the matrix of token embeddings $\mathbf{E}$ to produce a single sequence embedding $\mathbf{z} = \psi(\mathbf{E}) \in \mathbb{R}^d$ [55].

There are two aggregation strategies that are prevalent in practice. On the one hand, [CLS]-pooling extracts the representation of the special classification token prepended to the input:

$$\mathbf{z}_{[\text{CLS}]} = \mathbf{e}_{[\text{CLS}]} \quad (2.4)$$

While architecturally convenient, this representation is trained for next-sentence prediction in BERT or not explicitly trained at all in RoBERTa and has been shown to perform poorly on semantic similarity tasks without dedicated fine-tuning [35, 39, 56].

On the other hand, mean-pooling computes the average over all token representations

$$\mathbf{z}_{\text{mean}} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mathbf{e}_i \quad (2.5)$$

where $\mathcal{T} \subseteq \{1, \dots, n\}$ denotes the set of token indices. Reimers and Gurevych [35] empirically demonstrate that mean-pooling consistently outperforms [CLS]-pooling for encoders that have not been fine-tuned for sentence-level tasks.

However, even with mean-pooling, representations from vanilla pre-trained encoders yield sub-optimal sentence embeddings, since the aggregation is applied to representations that were never trained to encode sequence-level semantics [35, 56]. To address this, Reimers and Gurevych [35] proposed Sentence-BERT, which fine-tunes the encoder using a Siamese architecture with a contrastive objective on natural language inference labels, producing sequence embeddings $\mathbf{z}$ where cosine similarity directly reflects semantic relatedness. Gao et al. [57] later demonstrated with SimCSE that comparable quality can be achieved through simpler contrastive objectives, using dropout augmentation as a minimal positive-pair generator.

2.1.4 Similarity Measures

The ultimate goal of learning structured representation spaces is to enable the use of geometric operations to capture semantic relationships [32]. The most fundamental operation is the measurement of similarity or distance between two vectors $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{R}^d$. Common similarity measures include, the Euclidean distance $d(\mathbf{z}_1, \mathbf{z}_2) = \sqrt{\sum_{i=1}^d (z_{1i} - z_{2i})^2}$, which captures absolute distance in the vector space and the cosine similarity $\text{sim}(\mathbf{z}_1, \mathbf{z}_2) = \frac{\mathbf{z}_1^\top \mathbf{z}_2}{\|\mathbf{z}_1\| \|\mathbf{z}_2\|}$, which measures the angle between two vectors and is invariant to their magnitude [34, p. 103 f., p. 112 f.][43, p. 48 f.]. As illustrated in Figure 2.3, the choice of similarity measure can significantly impact downstream tasks such as retrieval, clustering, or classification, as it defines the geometric notion of semantic relatedness in the embedding space [35, 36].

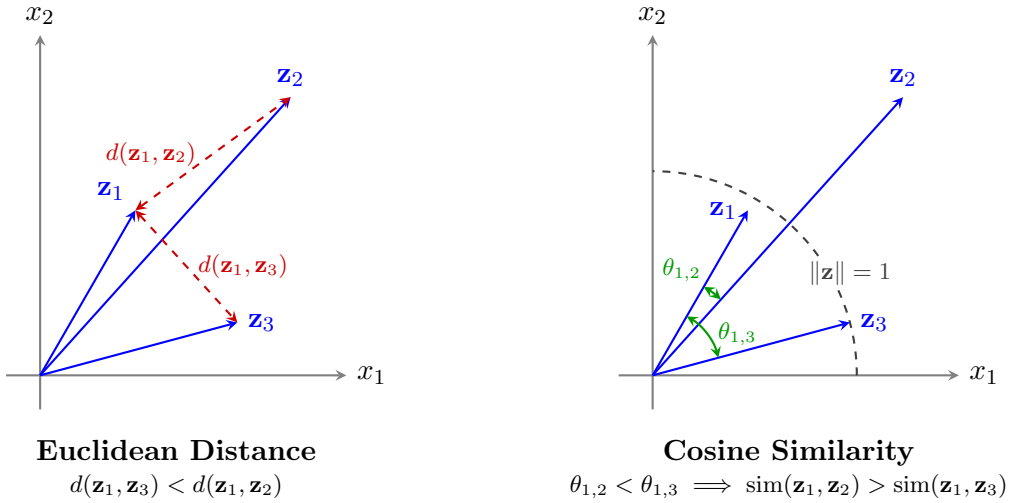


Figure 2.3: Geometric comparison of similarity measures. Depending on the chosen metric, the resulting similarity rankings diverge: In the Euclidean space (**left**), $\mathbf{z}_1$ is considered more similar to $\mathbf{z}_3$ due to their spatial proximity, as the metric is sensitive to vector magnitudes. Conversely, the cosine similarity (**right**) evaluates only the angular alignment (θ), identifying $\mathbf{z}_1$ and $\mathbf{z}_2$ as the most similar pair. (Own illustration)

2.2 The Geometry of High-Dimensional Representation Spaces

The utility of any vectorial representation ultimately depends on how efficiently the encoder exploits the available capacity of $\mathbb{R}^d$. A representation-space in which all vectors collapse into a narrow cone wastes capacity and a space in which a few points dominate the neighbourhood structure is structurally unstable. These qualities can be diagnosed purely from the intrinsic geometry of $\mathcal{Z}$, before any downstream task is even defined [21, 58].

That geometric regularity limits representational quality was recognised early across paradigms. Implicitly by Salton et al. [45] through angular discriminability in the term-document space, by Deerwester et al. [59] through the low effective rank revealed by SVD and by Mikolov et al. [32] through the presupposition that vector arithmetic carries semantic meaning. While originally articulated for frequency-based and static representations, these desiderata prove equally critical for modern contextual encoders, where recent studies demonstrate that Transformer-based embeddings also suffer from geometric pathologies [21, 60, 61].

2.2.1 Global Distributional Regularity: Isotropy

One of the most essential properties of a well-structured embedding space is that it should be isotropic, meaning that the variance of the representations is distributed uniformly across all

available dimensions. If variance is evenly distributed, no single spatial direction disproportionately dominates similarity computations, enabling the space to achieve maximum dynamic range to discriminate between nuanced semantic concepts [62, 63].

Definition 2.1 (Isotropy). *A set of embeddings $\{\mathbf{z}_1, \dots, \mathbf{z}_N\} \in \mathcal{Z} \subset \mathbb{R}^d$ with centroid $\boldsymbol{\mu} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i$ is called isotropic if its covariance matrix $\boldsymbol{\Sigma}$ is proportional to the identity matrix $\mathbf{I}_d$:*

$$\boldsymbol{\Sigma} = \frac{1}{N} \sum_{i=1}^N (\mathbf{z}_i - \boldsymbol{\mu})(\mathbf{z}_i - \boldsymbol{\mu})^\top = \sigma^2 \mathbf{I}_d \quad (2.6)$$

The eigendecomposition $\boldsymbol{\Sigma} = \mathbf{Q} \text{diag}(\lambda_1, \dots, \lambda_d) \mathbf{Q}^\top$ of a perfectly isotropic space yields identical eigenvalues $\lambda_1 = \dots = \lambda_d = \sigma^2$. Consequently, the expected cosine similarity between two randomly drawn vectors converges to zero as $d \rightarrow \infty$ [62, 63].

In practice, however, contextual representations are frequently highly anisotropic. They tend to cluster within a narrow cone in $\mathbb{R}^d$, leaving most of the space unused [21, 58]. To understand the consequences of this degeneration, any embedding can be decomposed into the global corpus centroid and a specific semantic residual component, $\mathbf{z}_i = \boldsymbol{\mu} + \tilde{\mathbf{z}}_i$. For L_2 -normalised vectors, the inner product then expands as

$$\mathbf{z}_i^\top \mathbf{z}_j = \underbrace{\|\boldsymbol{\mu}\|_2^2}_{\text{constant bias}} + \underbrace{\boldsymbol{\mu}^\top \tilde{\mathbf{z}}_j + \tilde{\mathbf{z}}_i^\top \boldsymbol{\mu}}_{\text{mean-residual interaction}} + \underbrace{\tilde{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_j}_{\text{semantic signal}}. \quad (2.7)$$

The term $\|\boldsymbol{\mu}\|_2^2$ acts as a constant bias for all vector pairs. When this term is large relative to the actual semantic residual $\tilde{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_j$, it overshadows every comparison, so that, all vectors appear artificially similar to one another⁵ [21, 60].

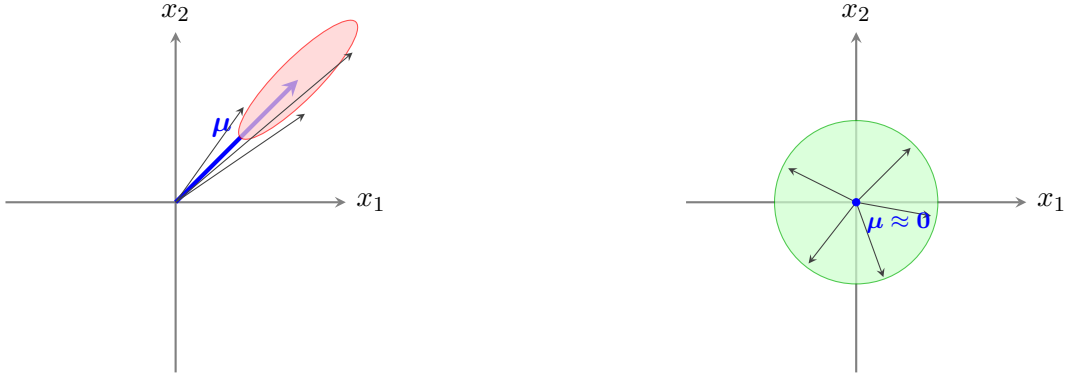


Figure 2.4: Visual comparison of representation space geometry. **Left:** A pathological, anisotropic space, where embeddings form a narrow cone shifted away from the origin. **Right:** An ideal, isotropic space where the full dynamic range of $\mathbb{R}^d$ is utilized for semantic discrimination, with a centered centroid. (Own illustration)

To systematically assess the extent of this geometric degeneration, visualized in Figure 2.4, several metrics have been proposed. The most direct and intuitive symptom is a high baseline angular similarity between random representations [21, 22, 62, 64].

⁵In a theoretical setting with perfect numerical precision, this constant bias would not affect the downstream ranking, as adding a constant to all similarities preserves their relative order. In practice, however, modern hardware relies on limited floating-point arithmetic. When the semantic variance is infinitesimally small compared to the massively dominant constant bias, it is frequently lost to rounding errors, effectively destroying the retrieval signal [58].

Metric: Average Pairwise Cosine Similarity $\bar{s}_{\cos}$ [21]

$$\bar{s}_{\cos} = \mathbb{E}_{i \neq j}[\cos(\mathbf{z}_i, \mathbf{z}_j)] = \binom{N}{2}^{-1} \sum_{i < j} \frac{\mathbf{z}_i^\top \mathbf{z}_j}{\|\mathbf{z}_i\|_2 \|\mathbf{z}_j\|_2} \quad (2.8)$$

Ideal: ≈ 0 (diverse directions) **Pathological:** $\rightarrow 1$ (cone-shaped concentration)

Isolating the primary cause of this baseline similarity compression requires examining the global centroid of the representations [65]. Mu and Viswanath [60] and Ethayarajh [21] originally demonstrated that contextual embeddings share a massively dominant common vector. Building upon these observations, Yamagiwa and Shimodaira [65] formalized the phenomenon by mathematically decomposing the mean squared norm of contextualized embeddings into their true semantic variance and the squared norm of their mean embedding. Following their framework, the exact influence of this spatial shift can be strictly quantified by the mean dominance ratio D_μ , which measures the proportion of the vector magnitude that is merely a shared spatial artifact rather than unique semantic information.

Metric: Mean Dominance Ratio D_μ [65]

$$D_\mu = \frac{\|\boldsymbol{\mu}\|_2^2}{\frac{1}{N} \sum_{i=1}^N \|\mathbf{z}_i\|_2^2} \quad (2.9)$$

Ideal: 0 (centroid at origin) **Pathological:** $\rightarrow 1$ (constant bias dominates)

While D_μ captures the translational shift away from the origin, it does not describe the shape of the variance. Moving beyond simple mean-shifts, the eigenvalue spectrum of the covariance matrix provides a comprehensive view of how the remaining variance is distributed and IsoScore quantifies how uniformly this variance is distributed across the embedding dimensions [62].

Metric: IsoScore [62]

Let $\lambda_1 \geq \dots \geq \lambda_d$ be the eigenvalues of $\boldsymbol{\Sigma}$ and $\bar{\lambda}$ their mean.

$$\text{IsoScore} = 1 - \frac{d \sum_{i=1}^d (\lambda_i - \bar{\lambda})^2}{(d-1) \left(\sum_{i=1}^d \lambda_i \right)^2} \quad (2.10)$$

Ideal: 1 (all eigenvalues equal) **Pathological:** $\rightarrow 0$ (single dominant direction)

An alternative, non-spectral approach probes the spatial density of the embedding space by evaluating its response to arbitrary directional unit vectors. This partition function evaluates whether the embeddings densely populate all directions or cluster along a few specific axes [60, 66].

Metric: Partition Function Isotropy I_{PF} [60, 66]

For a random unit vector $\mathbf{c} \in \mathbb{S}^{d-1}$, define the partition function $Z(\mathbf{c}) = \frac{1}{N} \sum_{j=1}^N \exp(\mathbf{z}_j^\top \mathbf{c})$. Isotropy is then defined as the ratio of the minimum to the maximum response:

$$I_{\text{PF}} = \frac{\min_{\mathbf{c}} Z(\mathbf{c})}{\max_{\mathbf{c}} Z(\mathbf{c})} \quad (2.11)$$

Ideal: 1 (Z constant across all directions) **Pathological:** $\rightarrow 0$ (strong directional dependence)

Finally, the simplest spectral diagnostic bounds the variance imbalance by comparing the ex-

treme ends of the eigenvalue spectrum. This condition number serves as a direct indicator of how strongly the variance collapses into the first few principal components [21, 62][67, p. 88 f.].

Metric: Eigenvalue Ratio R_λ [21, 62]

The condition number of the covariance matrix, expressed as the ratio of its largest to smallest eigenvalue:

$$R_\lambda = \frac{\lambda_1}{\lambda_d} \quad (2.12)$$

Ideal: 1. **Pathological:** $\gg 1$ (variance concentration along the first Principal Components (PCs)).

2.2.2 Information Density and Intrinsic Dimensionality

Although embeddings reside in a high-dimensional space $\mathbb{R}^d$ e.g., $d = 768$ or $d = 3072$, the actual data typically concentrate on a manifold of significantly lower dimensionality [68]. The intrinsic dimensionality d_{int} measures the minimum number of degrees of freedom required to capture the meaningful semantic variance of the dataset, treating the remaining dimensions as negligible statistical noise [69]. If d_{int} remains high relative to the ambient dimension d , the representation space uses its capacity efficiently. Conversely, a very low relative intrinsic dimension indicates a highly redundant encoding where semantic nuance is compressed into a few dominant components. To operationalise this manifold concept for empirical evaluation, global spectral estimators quantify this effective dimensionality continuously through the eigenvalue spectrum of the covariance matrix [69, 70]. A measure of the effective rank that represents the number of equally contributing dimensions is the Participation Ratio (PR) [70].

Metric: Participation Ratio (PR) [70]

$$\text{PR} = \frac{\left(\sum_{i=1}^d \lambda_i\right)^2}{\sum_{i=1}^d \lambda_i^2} \quad (2.13)$$

Ideal: $\text{PR} \approx d$ (perfect isotropy) **Pathological:** $\text{PR} \rightarrow 1$ (variance collapses into a single dimension)

While PR uses the sum of the squares of the eigenvalues, the effective rank is based on the Shannon entropy of the eigenvalue distribution. It provides a more precise measure of how many dimensions effectively carry information [64].

Metric: Effective Rank [64]

Let $p_i = \lambda_i / \sum_{j=1}^d \lambda_j$ be the normalized eigenvalues, forming a probability distribution. The effective rank is the exponential of the Shannon entropy H :

$$\text{Effective Rank} = \exp \left(- \sum_{i=1}^d p_i \ln p_i \right) \quad (2.14)$$

Ideal: $\approx d$ (uniform variance distribution) **Pathological:** 1 (all variance in one dimension)

For a more discrete threshold, the cumulative explained variance determines the minimal subspace dimensionality capturing the majority of the signal [70][71, p. 325 f.].

Metric: PCA Explained Variance Threshold d_α [70, 71]

The smallest number of principal components k required to explain a fixed threshold α of the total variance:

$$d_\alpha = \min \left\{ k \mid \frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^d \lambda_i} \geq \alpha \right\} \quad (2.15)$$

Ideal: High d_α relative to d **Pathological:** $d_\alpha \ll d$

While spectral methods capture global structure, they may miss locally varying dimensionality. Local estimators instead exploit the scaling behaviour of Euclidean distances in the immediate neighbourhood of each point [72].

Metric: Maximum Likelihood Estimator (MLE) [72]

For a point $\mathbf{z}$ with k -nearest-neighbour distances $r_1 \leq \dots \leq r_k$, the local topological dimension is estimated as:

$$\hat{d}_{\text{MLE}}(\mathbf{z}) = \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_k}{r_j} \right]^{-1} \quad (2.16)$$

The corpus-level estimate is obtained by averaging over all points.

Ideal: High $\hat{d}_{\text{MLE}}$ **Pathological:** $\hat{d}_{\text{MLE}} \rightarrow 1$

MLE is often sensitive to fluctuations in density. The Two Nearest Neighbors (TwoNN) estimator offers a more robust local alternative by considering only the ratio of distances to the first and second nearest neighbors, which makes it invariant to local differences in density [73].

Metric: TwoNN Estimator [73]

Let r_1 and r_2 be the distances to the first and second nearest neighbors. The distance ratio is defined as $\mu = r_2/r_1 \geq 1$. In a space of intrinsic dimension d , the cumulative distribution function of μ is $F(\mu) = 1 - \mu^{-d}$. The intrinsic dimension is estimated by finding the slope d of the linear relationship:

$$\log(1 - F(\mu)) = -d \log(\mu) \quad (2.17)$$

Ideal: High d relative to ambient space **Pathological:** $d \rightarrow 1$

2.2.3 Local Topology: Alignment and Uniformity

Wang and Isola [22] decompose the quality of representations on the unit hypersphere $\mathbb{S}^{d-1}$ into two converse properties. These form the theoretical foundation of modern Contrastive Learning frameworks such as SimCSE, which explicitly optimise the InfoNCE loss to simultaneously maximise alignment and uniformity [57]. When ground-truth semantic relationships are known, alignment measures how robustly the encoder maps related inputs to adjacent points [22].

Metric: Alignment [22]

Given a distribution of semantically related (positive) pairs p_{pos} , alignment is defined as the expected distance between them:

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{(x, x^+) \sim p_{\text{pos}}} [\|\phi_\theta(x) - \phi_\theta(x^+)\|_2^2] \quad (2.18)$$

Ideal: Low $\mathcal{L}_{\text{align}}$ (positive pairs are mapped closely) **Pathological:** High $\mathcal{L}_{\text{align}}$

Opposite to alignment, uniformity evaluates the overall spatial distribution of the representations without requiring labelled pairs. A perfectly isotropic representation projected onto $\mathbb{S}^{d-1}$ achieves high uniformity [22].

Metric: Uniformity [22]

Uniformity measures how evenly the marginal distribution of all representations covers the hypersphere:

$$\mathcal{L}_{\text{uniform}} = \log \mathbb{E}_{x, y \sim p_{\text{data}}} [e^{-t \|\phi_{\theta}(x) - \phi_{\theta}(y)\|_2^2}], \quad t > 0 \quad (2.19)$$

Ideal: Low $\mathcal{L}_{\text{uniform}}$ (vectors are spread evenly) **Pathological:** High $\mathcal{L}_{\text{uniform}}$ (vectors form degenerate clusters)

2.2.4 Spatial Fairness: Hubness

The curse of dimensionality introduces distortions into the neighbourhood structure of vector spaces. By the law of large numbers, as the dimension $d \rightarrow \infty$, the pairwise Euclidean distance between any two independent and identically distributed vectors converges to a constant [31, p. 172 f.][74]. Consequently, the relative contrast between the distance of the nearest $D_{\min}$ and farthest $D_{\max}$ neighbour of any arbitrary reference point vanishes by $\lim_{d \rightarrow \infty} \frac{D_{\max} - D_{\min}}{D_{\min}} = 0$ [74]. A direct consequence of this distance concentration is the emergence of hubness. In finite datasets drawn from such high-dimensional distributions, points situated closer to the global data mean act as *hubs*. They appear as the nearest neighbours to a disproportionately large number of other points. On the other hand, points near the spatial boundary become *anti-hubs* that act as nearest neighbours to no other points. To quantify this structural asymmetry, we analyse the distribution of nearest-neighbour occurrences across the dataset [75–77].

Metric: Hubness Skewness S_k [75, 76]

Let $N_k(i)$ be the number of times vector $\mathbf{z}_i$ appears as a k -nearest neighbor of all other points. Hubness is measured by the third standardised moment (skewness) of this distribution:

$$S_k = \frac{\mathbb{E}[(N_k - \mu_{N_k})^3]}{\sigma_{N_k}^3} \quad (2.20)$$

Ideal: $S_k \approx 0$ (symmetric, hub-free neighbourhood structure) **Pathological:** $S_k \gg 1$ (emergence of influential hubs)

Hubness is actively reinforced by low intrinsic dimensionality and anisotropy. This is because the irrelevant dimensions act as noise that inflates distances, pushing all points closer together in the relevant subspace [75, 76]. While skewness measures the asymmetry of the distribution, the fractions express the problem in terms of absolute ratios.

Metric: Hub and Anti-Hub Fraction [75, 76]

Let $N_k(i)$ be the number of times vector $\mathbf{z}_i$ appears as a k -nearest neighbor of all other points. The expected value under perfect spatial uniformity is $\mathbb{E}[N_k] = k$.

$$f_{\text{antihub}} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(N_k(i) = 0) \quad (2.21)$$

$$f_{\text{hub}} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(N_k(i) \geq 2k) \quad (2.22)$$

where $\mathbb{I}(\cdot)$ is the indicator function and $2k$ serves as a standard empirical threshold for hub emergence [76].

Ideal: 0 (all points are retrieved roughly k times) **Pathological:** $\rightarrow 1$

To capture the global inequality of the N_k distribution in a single number, economists often use the Robin Hood Index, also known as the Hoover Index. This index measures the proportion

of *nearest-neighbor slots* that would need to be redistributed from the hubs to the anti-hubs in order to achieve perfect equality [77].

Metric: Robin Hood Index [77]

Derived from the Gini coefficient framework, the index computes the maximum vertical distance between the Lorenz curve of the N_k distribution and the line of perfect equality:

$$\text{RHI} = \frac{1}{2} \frac{\sum_{i=1}^N |N_k(i) - k|}{\sum_{i=1}^N N_k(i)} \quad (2.23)$$

Ideal: 0 (perfect equality, $N_k(i) = k$ for all) **Pathological:** $\rightarrow 1$ (maximum inequality)

This list of geometric properties is not exhaustive. For instance, additional topological properties such as local clustering coefficients or manifold curvature could be considered to capture more nuanced spatial pathologies. However, the metrics formalised above are fundamentally sufficient to capture the most critical geometric pathologies that degrade downstream retrieval performance.

2.3 Retrieval-Augmented Generation

Historically, vector embeddings were primarily utilized for discriminative NLP tasks, such as text classification or clustering [32, 34, 39]. However, with the recent paradigm shift towards GenAI, LLMs have become the focal point of modern NLP [1].

These generative models fundamentally require a mechanism for accessing and utilizing factual knowledge to produce informed responses. This knowledge acquisition is typically classified into the two paradigms parametric and non-parametric memory [11]. Parametric knowledge is implicitly learned within the model’s internal weights θ during training [9]. To update this knowledge or adapt the model to a specialized domain, the model must undergo continuous training or fine-tuning. However, this parametric approach is computationally expensive, inflexible and prone to hallucination [8].

On the other hand, non-parametric memory explicitly provides knowledge within the model’s input context. RAG operates on this second paradigm by dynamically retrieving relevant texts from an external knowledge base and integrating them into the generation process visualized in Figure 2.5. As Barnett et al. [10] note, RAG provides a robust alternative to fine-tuning by relying on LLMs exclusively for the synthesis of answers based on explicitly retrieved knowledge. This allows not only to mitigate hallucinations but also to access up-to-date information without retraining [11, 27, 78].

The core idea of retrieving relevant documents to satisfy an information need is not new. It is a fundamental problem that has been extensively studied in the field of Information Retrieval (IR) [36]. At an abstract level, RAG utilizes the same workflow as classical IR system. An offline indexing phase, in which the knowledge corpus $\mathcal{C}$ is processed and structured for efficient retrieval and an online phase, in which the system dynamically fetches relevant information to ground the generated response for a user query q [27]. The primary retrieval objective remains identical to classical IR, identifying a subset of documents $\mathcal{C}_{q,k}$ from a massive corpus $\mathcal{C}$ that optimally satisfies the given information need q . However, unlike traditional IR where the end-user is human, the *consumer* of the retrieved documents in a RAG system is an LLM. This paradigm shift critically alters the retrieval requirements [27]. While human users can easily skim a list of search results and ignore irrelevant hits, generative models possess finite context windows and are highly susceptible to reasoning errors when distracted by noisy information [79]. Consequently, RAG retrieval must prioritize precision over broad recall, strictly tailoring the fetched context to the technical constraints of the downstream LLM [27, 80].

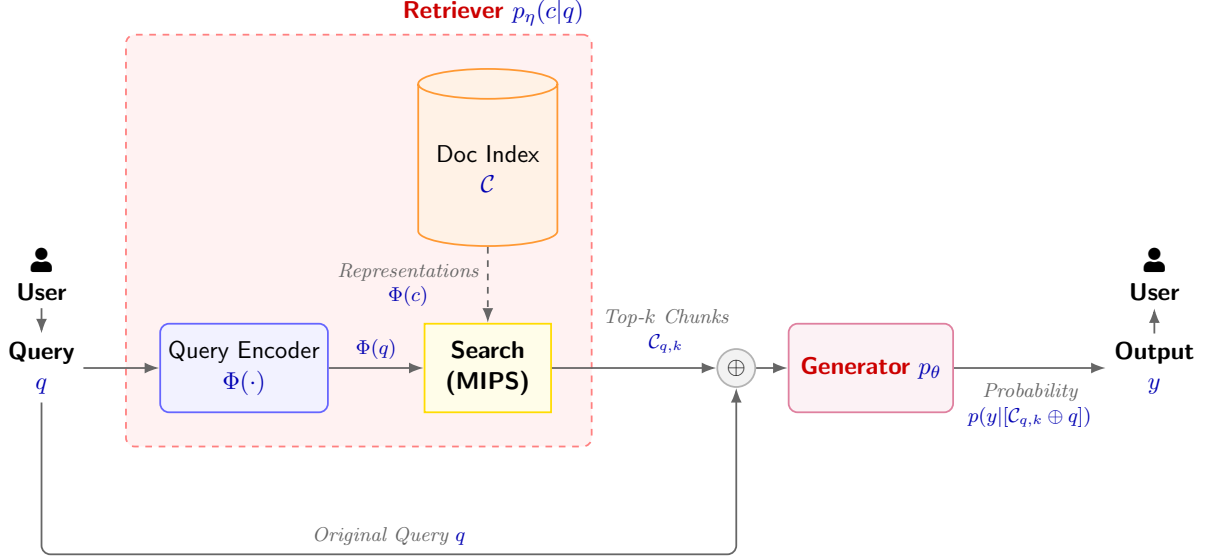


Figure 2.5: Core architecture of a modern standard In-Context RAG system. Retrieved documents ($\mathcal{C}_{q,k}$) are explicitly concatenated ($\oplus$) with the original query (q) and passed to the generator as an augmented context window. (Own Illustration)

2.3.1 Formalizing the RAG Paradigm

Lewis et al. [11] initially formalized the theoretical foundation of the RAG framework⁶. The user's request to the LLM, the prompt, is referred to as the query $q \in \mathcal{Q} \subset \mathcal{X}$. The external knowledge corpus $\mathcal{C}_{\text{info}}$ is segmented into discrete chunks, with each chunk representing a discrete information unit $c \in \mathcal{C}$ [27, 81]. Fundamentally, a RAG-model is actually a composition, or orchestration, of two distinct probabilistic models:

1. **The Retriever** $p_\eta(c | q)$: A model parameterized by η that returns a probability distribution over text chunks $c \in \mathcal{C}$ given the query q . In practice, this returns a top- k truncated set of candidates, denoted as $\mathcal{C}_{q,k}$ [11, 81].
2. **The Generator** $p_\theta(y_i | q, c, y_{1:i-1})$: An autoregressive LLM parameterized by θ that computes the probability of generating the next token y_i conditioned on the original query q , one specific retrieved chunk c and the previously generated tokens $y_{1:i-1}$ [11].

The probability of generating the full target sequence $y = (y_1, \dots, y_m)$ is the product of these individual token probabilities. Lewis et al. [11] introduced two primary methods to combine the retriever and the generator, treating the retrieved chunk c as a latent variable that must be marginalized out over the candidate set $\mathcal{C}_{q,k}$.

RAG-Sequence Model. This model assumes that a single retrieved chunk c provides the context for generating the entire target sequence y . Therefore, the total probability is the sum over all retrieved chunks [11]:

$$\mathbb{P}_{\text{RAG-Seq}}(y | q) = \sum_{c \in \mathcal{C}_{q,k}} p_\eta(c | q) \prod_{i=1}^m p_\theta(y_i | q, c, y_{1:i-1}) \quad (2.24)$$

⁶To ensure consistency with notation established in Section 2.1, their original notation will be adjusted here.

RAG-Token Model. This model allows the generator to draw upon a different chunk for every single token it produces, marginalizing the latent variable c at the token level rather than the sequence level [11]:

$$\mathbb{P}_{\text{RAG-Token}}(y \mid q) = \prod_{i=1}^m \sum_{c \in \mathcal{C}_{q,k}} p_{\eta}(c \mid q) p_{\theta}(y_i \mid q, c, y_{1:i-1}) \quad (2.25)$$

2.3.2 Modern In-Context RAG

Although this first formalization of RAG is a robust probabilistic framework, it is computationally expensive and complex to implement, especially during inference. The need to compute probabilities over multiple retrieved chunks at every generation step leads to significant overhead. As a result, the contemporary standard for RAG, often referred to as *Standard RAG* or *In-Context RAG* or *Naive RAG*, has evolved to simplify this process by integrating the retrieved knowledge directly into the prompt without explicit marginalization [27]. These approaches concatenate the top- k retrieved chunks into a single context string $[\mathcal{C}_{q,k} \oplus q]$. The generation probability

$$\mathbb{P}_{\text{In-Context}}(y \mid q, \mathcal{C}_{q,k}) = \prod_{i=1}^m p_{\theta}(y_i \mid [\mathcal{C}_{q,k} \oplus q], y_{1:i-1}) \quad (2.26)$$

collapses into a standard autoregressive language modeling objective, conditioned directly on the prompt [82]. This simplification clearly shows that the quality of the generative output depends largely on the relevance and quality of the candidate set $\mathcal{C}_{q,k}$. If the retriever $p_{\eta}(c \mid q)$ is unable to find the correct information, the generator p_{θ} cannot compensate for this effectively, which identifies the retrieval process as a significant bottleneck [10, 80, 83].

2.3.3 The Retrieval Engine and Advanced Pipelines

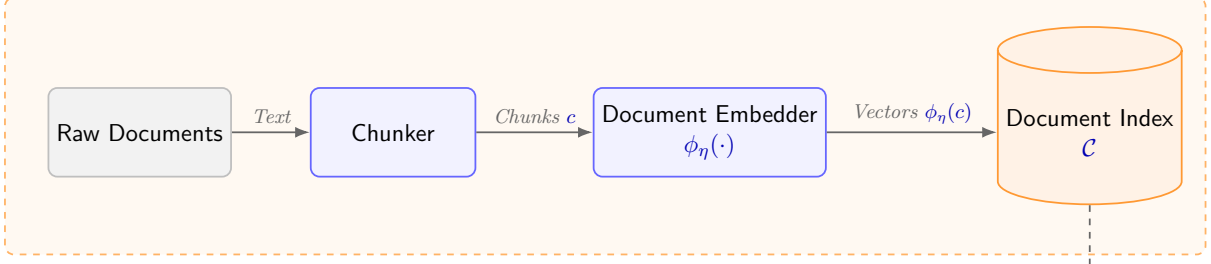
To satisfy this strict precision constraints imposed by the downstream generator, the retrieval engine p_{η} is typically instantiated as a dense bi-encoder [81]. Connecting back to the embedding theory established in Section 2.1, the retriever leverages a parameterized embedding function ϕ_{η} to independently map both the user query $q \in \mathcal{Q}$ and all corpus chunks $c \in \mathcal{C}$ into the shared continuous representation space $\mathcal{Z}$. The retrieval probability of a chunk $p_{\eta}(c \mid q)$ is then derived by applying a softmax function over the geometric similarities between these vectors, typically computed via Maximum Inner Product Search (MIPS). Thus, the likelihood of retrieving a specific chunk is proportional to the exponential of its cosine similarity to the query:

$$p_{\eta}(c \mid q) \propto \exp(\text{sim}(\phi_{\eta}(q), \phi_{\eta}(c))) \quad (2.27)$$

Given the critical role of this geometric retrieval step in the overall RAG performance, various architectural optimisations have been proposed to enhance both retrieval quality and computational efficiency. As illustrated in Figure 2.6, such *Advanced RAG* systems extend the basic paradigm by introducing specialized pre- and post-processing modules to refine the information flow before and after the vector search [10, 27].

The pre-retrieval process optimises both the document index and the user query prior to the vector search. Indexing quality is enhanced by balancing chunk granularity and appending metadata for structural filtering [27]. Meanwhile, initial user queries q are often ambiguous or incomplete. They are refined via rewriting, expansion, or transformation into an optimised representation q' that is better suited for semantic retrieval [27, 84–86].

Offline Indexing Process



Online Generation & Query Process

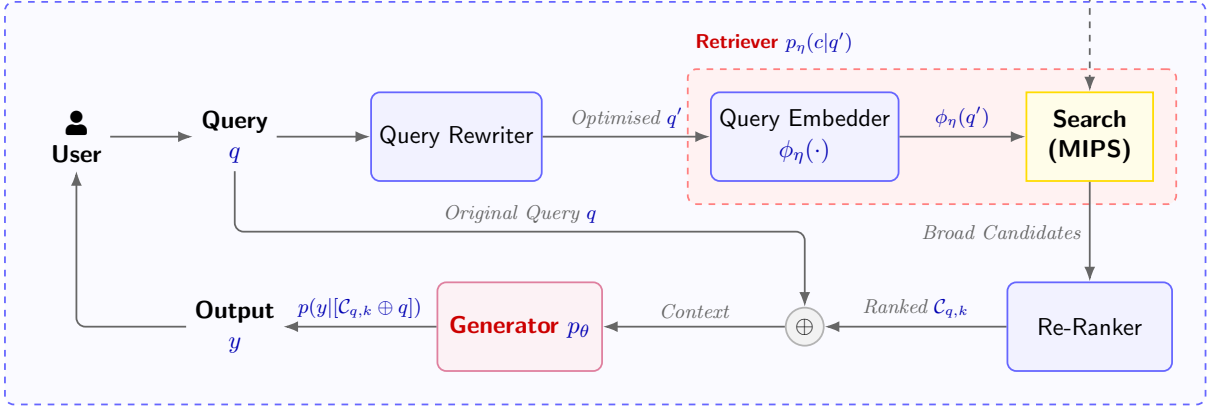


Figure 2.6: The Advanced RAG Pipeline incorporating query rewriting for vector optimisation and cross-encoder re-ranking for precise context extraction. (Own illustration in adaptation of [27])

The post-retrieval process curates the retrieved candidate set $\mathcal{C}_{q,k}$ before generation. Feeding uncurated documents directly into LLMs causes information overload and exacerbates the *lost-in-the-middle* effect, where models ignore information buried in the center of the prompt [27, 79]. To mitigate this, context compression and reranking are applied. Reranking employs computationally heavier Cross-Encoders to re-evaluate the broad candidate set, prioritizing and relocating the most relevant documents to the edges of the context window, where the generator’s attention peaks [87].

Beyond these sequential pre- and post-processing steps, the design space has recently evolved towards *Modular RAG* architectures. This paradigm transcends fixed, chain-like pipelines by introducing flexible, interchangeable modules [27]. Approaches such as dynamic routing across diverse data sources [88], multi-hop retrieval for complex reasoning [89] and active relevance feedback loops allow the system to adaptively control the retrieval flow based on the specific query context, further extending the capabilities and robustness of modern RAG architectures [27, 90, 91].

2.3.4 Evaluation of RAG Systems

As the modular architecture of RAG suggests, system evaluation ranges from holistic end-to-end perspectives to granular component-level assessments. Holistic approaches evaluate the final generated response y , often employing traditional reference-based string-matching metrics such as BLEU and ROUGE, or modern LLM-as-a-judge frameworks [27, 92, 93]. However, these end-to-end measures fail to provide actionable insights into the specific contributions of the underlying subsystems [27]. To effectively diagnose the root cause of a system failure, distinguishing whether the generator p_θ hallucinated despite being provided with correct context, or whether the retriever p_η fundamentally failed to find the necessary documents, evaluation

frameworks must strictly decouple generation quality from retrieval quality [10, 93]. Operating under the premise that accurate generation is strictly bounded by accurate retrieval, this thesis focuses entirely on evaluating the retrieval engine.

As established, the retrieval step can be framed as a classical IR problem, allowing for the direct application of IR-ranking metrics to the top- k candidate set $\mathcal{C}_{q,k}$. With $\mathcal{Q}$ denoting the set of all queries and $\mathcal{R}_q$ representing the set of ground-truth relevant documents for a specific query $q \in \mathcal{Q}$, one of the simplest measurements is Precision@ k . It captures the fraction of the retrieved top- k relevant documents⁷[36, p. 155][92].

Metric: Precision@ k

$$\text{Precision@}k = \frac{|\mathcal{C}_{q,k} \cap \mathcal{R}_q|}{k} \quad (2.28)$$

Ideal: 1 (All top- k documents are relevant). **Pathological:** 0.

Another foundational metric is Recall@ k , which measures the fraction of all ground-truth relevant documents that are successfully retrieved within the top- k candidates [36, p. 155].

Metric: Recall@ k

$$\text{Recall@}k = \frac{|\mathcal{C}_{q,k} \cap \mathcal{R}_q|}{|\mathcal{R}_q|} \quad (2.29)$$

Ideal: 1 (All relevant documents are found). **Pathological:** 0.

Both Precision and Recall are set-based measures. They evaluate only the presence or absence of documents, disregarding their specific order. While these methods are foundational, they are insufficient for evaluating RAG systems [36, p. 158 f.][93]. In a RAG context, the order of documents directly impacts the downstream generator, as LLMs are highly susceptible to *lost-in-the-middle* effects [79]. Thus, evaluation requires rank-aware metrics that penalize systems for placing relevant documents lower in the context window [36, p. 159]. To apply these rank-aware metrics, the retrieved candidate set $\mathcal{C}_{q,k}$ is formalized as an ordered sequence, or to be more precise a ranked list $L_{q,k} = (c_1, c_2, \dots, c_k)$. One widely used rank-aware metric is the Mean Reciprocal Rank (MRR), which evaluates the position of the first relevant document within this ordered list [92].

Metric: Mean Reciprocal Rank (MRR)

Let rank_i be the position of the first relevant document for the i -th query in the ranked list $L_{q,k}$. The MRR averages the reciprocal of this rank across all $|\mathcal{Q}|$ queries:

$$\text{MRR} = \frac{1}{|\mathcal{Q}|} \sum_{i=1}^{|\mathcal{Q}|} \frac{1}{\text{rank}_i} \quad (2.30)$$

Ideal: 1 (The first retrieved document is always relevant). **Pathological:** $\rightarrow 0$.

Unlike Recall and MRR, which assume binary relevance, the Normalized Discounted Cumulative Gain (nDCG) accommodates graded relevance [36, p. 162 f.][92].

⁷Crucially, the definition of this relevant set $\mathcal{R}_q$ is highly domain-dependent. As Van Opijnen and Santos [15] emphasize for Legal Information Retrieval (LIR), relevance extends far beyond simple topical overlap. In legal and tax contexts, a document’s relevance is multidimensional, incorporating hierarchical authority e.g., a high-court ruling outweighs a lower-court ruling, temporal validity and dense cross-referential relationships [15]. Therefore, evaluating retrieval in specialized domains requires a preliminary definition of relevance.

Metric: Normalized Discounted Cumulative Gain (nDCG@k)

Let rel_i be the graded relevance score of the document at rank i in $L_{q,k}$. The DCG penalizes relevant documents that appear lower in the ranking using a logarithmic discount:

$$DCG@k = \sum_{i=1}^k \frac{rel_i}{\log_2(i+1)} \quad (2.31)$$

To ensure comparability across queries with varying numbers of relevant documents, the score is normalized by the Ideal DCG (IDCG@k), which is the DCG obtained for an perfectly sorted list $\hat{L}_{q,k}$.

$$nDCG@k = \frac{DCG@k}{IDCG@k} \quad (2.32)$$

Ideal: 1 (Perfect ranking of all relevant documents). **Pathological:** 0.

In practice, the choice of the primary evaluation metric must align with the specific requirements of the downstream RAG application [27, 92].

2.4 Embedding Optimisation Strategies

As established in Section 2.2, embedding models frequently produce pathologically structured representation spaces. While pipeline-level interventions within a RAG framework (cf. Section 2.3.3) attempt to compensate for poor retrieval outcomes a posteriori, they do not resolve the underlying geometric distortions. To fundamentally heal the representation space, interventions must target the embeddings directly [21, 56, 58].

A highly efficient approach is the application of post-hoc geometric transformations. These methods apply parameter-free or computationally inexpensive linear-algebraic operations directly to the output embeddings $\mathbf{z} \in \mathbb{R}^d$ to actively enforce the geometric goals [94]. To systematically optimise this space, multiple mathematical angles exist, ranging from simple spatial translation to strict variance equalization [60, 95].

2.4.1 Centering and L_2 -Normalization

The most fundamental geometric corrections address the position and scale of the embedding space. As derived in Equation 2.7, a spatial shift away from the origin introduces a massive constant bias $\|\boldsymbol{\mu}\|_2^2$ that dominates similarity calculations. Centering, or Mean-Shift Removal, directly eliminates this bias by subtracting the corpus centroid $\boldsymbol{\mu}$ from every vector $\tilde{\mathbf{z}} = \mathbf{z} - \boldsymbol{\mu}$. This translational shift forces the mean dominance ratio to zero ($D_\mu \rightarrow 0$), ensuring that subsequent dot products exclusively measure the semantic residual signal [21, 60].

Following translation, L_2 -Normalization standardizes the scale of the vectors by dividing each embedding by its Euclidean norm $\hat{\mathbf{z}} = \tilde{\mathbf{z}}/\|\tilde{\mathbf{z}}\|_2$. This projects all representations onto the unit hypersphere $\mathbb{S}^{d-1}$ [22].

2.4.2 Dimensionality Reduction via Principal Component Analysis (PCA)

If the data manifold exhibits a low intrinsic dimensionality $d_{\text{int}} \ll d$ (cf. Section 2.2.2), the $d - d_{\text{int}}$ residual dimensions contribute statistical noise rather than semantic signal. PCA can be employed to compress the representations while retaining the primary semantic variance. The projection onto the leading d' principal components is formalized as:

$$\hat{\mathbf{z}} = \mathbf{P}_{d'}^\top (\mathbf{z} - \boldsymbol{\mu}) \in \mathbb{R}^{d'} \quad (2.33)$$

where $\mathbf{P}_{d'} = [\mathbf{v}_1, \dots, \mathbf{v}_{d'}] \in \mathbb{R}^{d \times d'}$ contains the top- d' eigenvectors of the covariance matrix $\boldsymbol{\Sigma}$ [71, p. 317 f.][96]. Discarding these noise dimensions mitigates the curse of dimensionality, heavily reduces computational retrieval costs and counters the emergence of hubness (cf.

Section 2.2.4) [75]. Because this linear projection alters the vector magnitudes, the reduced representations can be subsequently L_2 -normalized $\tilde{\mathbf{z}} = \hat{\mathbf{z}}/\|\hat{\mathbf{z}}\|_2$, projecting them onto the new unit hypersphere $\mathbb{S}^{d'-1}$ to ensure easier similarity calculations [71, p. 321 f.][96].

2.4.3 Mean-Shift and Bias Removal: All-but-the-Top (ABTT)

While standard PCA successfully removes low-variance noise, it inherently preserves the dominant directions of variance [71, p. 317 f.]. Mu and Viswanath [60] observed that these leading principal components in contextual embeddings predominantly encode frequency-related artefacts, stop-words and a global directional bias rather than discriminative semantic meaning. To fight this, they proposed the ABTT transformation, which removes the projection onto the top- k eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ from the centered embeddings $\tilde{\mathbf{z}}$:

$$\hat{\mathbf{z}} = \tilde{\mathbf{z}} - \sum_{j=1}^k (\tilde{\mathbf{z}}^\top \mathbf{v}_j) \mathbf{v}_j \quad (2.34)$$

By eliminating these dominant directions (typically $k \leq 5$), ABTT actively reduces the eigenvalue ratio R_λ , allowing the subtle semantic differences encoded in the lower principal components to dictate the cosine similarity [56, 60].

2.4.4 Variance Equalization: Whitening

Although ABTT successfully eliminates the most extreme axes of variance, the remaining dimensions may still scale unequally [60]. A more complex approach to establish a balanced representation space is whitening, which equalizes variance across all dimensions simultaneously by enforcing the strict isotropy condition $\Sigma = \sigma^2 \mathbf{I}_d$ [62, 63, 95].

Given the eigendecomposition $\Sigma = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$, with orthogonal eigenvectors $\mathbf{Q}$ and eigenvalues $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_d)$, the Zero-phase Component Analysis (ZCA) whitening matrix is defined as:

$$\mathbf{W}_{\text{ZCA}} = \mathbf{Q} \text{diag}\left(\frac{1}{\sqrt{\lambda_1 + \epsilon}}, \dots, \frac{1}{\sqrt{\lambda_d + \epsilon}}\right) \mathbf{Q}^\top \quad (2.35)$$

where $\epsilon > 0$ is a small regularisation constant preventing the infinite amplification of near-zero eigenvalues. Once the whitening matrix is computed, the new representation $\hat{\mathbf{z}}_i$ is generated by applying this linear transformation directly to the mean-centered original vector $\hat{\mathbf{z}} = \mathbf{W}_{\text{ZCA}} \tilde{\mathbf{z}}$. This projection actively reshapes the vector such that the resulting space achieves a perfect IsoScore of 1.0 [95]. Unlike PCA-based whitening, ZCA whitening preserves the original coordinate axes in a least-squares sense [97]. However, this strict mathematical isotropy involves a trade-off: dimensions with small eigenvalues are amplified by the factor $1/\sqrt{\lambda_i + \epsilon}$, which may potentially boost irrelevant variance alongside the semantic signal [96, 98].

2.4.5 Composability of Transforms

Crucially, these transform families are not exclusive and can be composed sequentially to accentuate their benefits. For instance, ABTT followed by PCA first removes the mean-shift and dominant artefact directions and subsequently reduces the dimensionality in the cleaned space. The order in which these operations are applied is of great importance. If, for example, PCA is applied after the whitening transformation, it operates on a space that has already been variance-equalised, in which the notion of *top components* differs fundamentally from the original anisotropic space [95, 96].

3 Related Work

The integration of AI into the legal domain has rapidly evolved into a highly active research area. This chapter reviews the current state of the art to provide the necessary context and to identify the specific research gap addressed by this thesis.

3.1 Domain-Adapted Language Models

Domain adaptation has proven to be a major leverage point for enhancing the performance of embedding models in specialised fields [18]. At its core, this adaptation relies on fine-tuning, a form of transfer learning, which Jurafsky and Martin [34] define as the process of tailoring a pre-trained language model to a specific domain or task. Models are first trained on large, generic corpora to learn foundational linguistic structures and subsequently fine-tuned on domain-specific data to capture the unique characteristics of the target domain [18]. This approach makes it possible to bridge the gap between the distributions of the training corpora and the target domain data, which has been determined to be the cause of the poor performance of standard embedding methods [18–20, 99].

Therefore domain specific models have been developed across several fields. In the biomedical domain, BioBERT by Lee et al. [19] and SciBERT by Beltagy et al. [99] demonstrated the efficacy of domain-adaptive pre-training through further pre-training on domain corpora. Clinical BERT by Alsentzer et al. [100] extended this to clinical notes. In the financial domain, FinBERT by Yang et al. [101] adapted BERT for financial communications, while BloombergGPT by Wu et al. [102] later demonstrated domain adaptation at the generative model scale.

Within the legal domain, Chalkidis et al. [20] provided the most systematic investigation of domain adaptation strategies for BERT, testing three approaches. First using BERT out of the box, second, further fine-tuning on domain-specific corpora and third, training from scratch with a domain-specific corpus-vocabulary. Their resulting model family, LEGAL-BERT, was trained on 12 GB of diverse English legal text and consistently outperformed generic baselines on downstream legal tasks, such as text classification on EURLEX57K, judgment prediction on ECHR cases and named entity recognition on contracts.

Furthermore, Hua et al. [103] explored with LegalReLECTRA architectural adjustments. They combined legal and medical corpora with a REFORMER-based architecture to process passages of up to 8,092 tokens. This addressed a practical limitation of standard BERT models whose 512-token window is insufficient for lengthy legal documents. Evaluations have remained focused on classification and sequence-labelling tasks. Moreover, Niklaus et al. [104] created with MultiLegalPile a 689 GB multilingual legal corpus spanning 24 languages, which he then used to pre-train a series of legal language models that established new state-of-the-art results on LexGLUE and LEXTREME, two comprehensive legal NLP benchmarks [104–106]¹.

While MultiLegalPile includes German-language legal data from Swiss federal court rulings, it does not address the German tax law domain specifically, nor does it evaluate embedding representations for retrieval tasks.

The German Tax Law Domain is directly addressed by two recent lines of work. Wind et al. [5] presented SteuerLLM, a domain-adapted large language model with 28 billion parameters specifically for German tax law, trained on synthetically generated data from authentic examination material via a controlled RAG pipeline. Alongside the model, they released SteuerEx,

¹Due to the rapid proliferation of domain-adapted language models across multiple fields, an exhaustive enumeration of all available embedding architectures is impractical. Extensive reviews of legal-specific models are provided by Chalkidis et al. [105].

the first open benchmark for German tax law, comprising 115 questions drawn from undergraduate and graduate university tax law examinations across six core domains. SteuerLLM demonstrates that a domain-adapted model outperforms substantially larger general-purpose models on realistic tax law reasoning tasks, confirming that domain-specific adaptation yields measurable benefits in this domain [5]. The work of Luketina et al. [78] takes a complementary approach and compares RAG architectures with the fine-tuning of GPT-4o specifically for Austrian VAT law. Crucially, both Wind et al. [5] and Luketina et al. [78] address domain adaptation exclusively at the generative model level, optimising the LLM that formulates the final answer, while treating the retrieval component as a fixed, underlying pipeline element.

3.2 Legal Information Retrieval

3.2.1 Characteristics and Challenges

The process of legal research is an integral component of a legal professional’s daily workflow and the development of tool support for this process, known as Legal Information Retrieval (LIR), has been a cornerstone of AI & Law research [15][107, p. 21]. Van Opijnen and Santos [15] provide a foundational theoretical framework for LIR, which is defined as a specific subtype of IR that has been justified by a variety of characteristics particular to legal text data. These include among other things specialised terminology, extensive synonymy, dense cross-referencing and a diversity of document types² [15]. They designed a six-dimensional model of relevance in LIR that extends beyond simple topical matching which highlights the unique structure and interconnectedness of legal information objects [15].

Most research in LIR addresses challenges that arise from the specific domain characteristics [108–111]. A major trend has been to leverage NLP technologies to access deeper semantic layers of legal documents. For instance, Grabmair et al. [112] combined an ontology of intellectual property law with NLP and machine learning to improve conceptual legal information retrieval of vaccine injury decisions. Schweighofer and Geist [113] explored ontologies and relevance feedback methods for query expansion. Waltl [4] engineered a sophisticated NLP-based tool suite for legal experts that includes both basic and advanced legal information retrieval functionality using the UIMA type system. Empirical studies emphasize the practical importance of good legal search or retrieval. Lastres [3], surveying 190 law firm associates, found that up to 15 hours per week are spent conducting legal research, while user studies by Peoples [114] and Wiggers et al. [115] investigated how legal professionals interact with different search modalities and relevance criteria.

The general challenges of the legal domain are fully reflected in the German legal domain. As Landthaler [107] observes, German legal information retrieval constitutes a rarely investigated topic in research, despite the distinctive challenges it poses and more research on German AI & Law is required [4]. German tax law in particular is widely regarded as an very complex legal field.

3.2.2 Retrieval Approaches for German Legal Text

The initial approaches to LIR were largely based on traditional term-based vector space models such as TF-IDF and Okapi BM25, which capture lexical overlap but struggle to account for the pervasive synonymy and semantic complexity of legal language [115]. To supplement these purely term-based methods, early advanced approaches explored the extraction and exploitation of citations and cross-references within legal documents as a complementary retrieval signal [107, 116].

With the introduction of distributional semantic embedding-models, multiple studies have explored their application to legal text representation. For example, Sugathadasa et al. [117]

²A more detailed definition of the legal domain can be found in Section ??

compared TF-IDF with Paragraph Vectors (doc2vec) for legal document retrieval. Adebayo et al. [118] applied normalised word embedding vectors to IR and textual entailment tasks on the COLIEE competition dataset. Vo et al. [119] used word embeddings to suggest synonyms during technology-assisted legal review and Smywiński-Pohl et al. [120] constructed a legal dictionary for Polish law using word embeddings. Consistently, these studies demonstrate the superiority of embedding-based approaches over pure keyword matching in LIR-tasks. Landthaler [107] confirms this finding for German legal search specifically, showing that static word embeddings effectively capture synonymy relations for query expansion in German tax law. He also proposed the WE-DF vector space model as an implicit query expansion mechanism that eliminates the need for manually maintained thesaurian approach, similarly explored by Pickel [121] on the German Civil Code. However, those contributions operate entirely within the pre-Transformer paradigm of static, context-independent word embeddings, leaving modern contextualised document representations unexplored.

3.2.3 Evaluating Retrieval in Legal RAG Systems

Since finding the most relevant documents from a large corpus given a user query is independent of whether the retrieved documents are presented directly to a user or passed as context to a generative model, the requirements for the retriever component of an RAG application are identical to those in the classic LIR. Therefore, the domain-specific challenges identified in the LIR literature apply in full to the retrieval stage of a RAG system operating on legal text [15][107, p. 24].

Recently, efforts have been made to isolate and address this specific retrieval component of RAG systems in legal settings through dedicated evaluations. Pipitone and Alami [13] introduced LegalBench-RAG, the first benchmark explicitly designed to evaluate retrieval quality in legal RAG independently of the downstream generator. Building on this benchmark, Reuter et al. [12] identified a critical failure mode called Document-Level Retrieval Mismatch (DRM), where retrievers select chunks from entirely incorrect source documents due to high lexical and structural similarity within legal corpora, observing DRM rates exceeding 95% in standard RAG configurations. Their proposed mitigation, Summary-Augmented Chunking, enriches each text chunk with a document-level synthetic summary to preserve global context. While their work demonstrates the vulnerability of retrieval in legal domains, it is restricted to English-language data and focuses strictly on the pre-retrieval pipeline, rather than the representational deficiencies of the embedding models themselves³ [12].

Contrary, for the German legal domain, evaluation has traditionally relied on classical IR benchmarks rather than RAG-specific frameworks. Wrzałik and Krechel [24] introduced German Dataset for Legal Information Retrieval (GerDaLIR), the first large-scale German legal information retrieval dataset, comprising over 131,000 collection documents segmented into more than 3 million passages with approximately 123,000 queries derived from case law references. Their baseline experiments compare term-based methods, word-centroid similarity using word embeddings and neural reranking with German BERT and ELECTRA models fine-tuned on the dataset. However, it covers general German legal language, predominantly civil and administrative law, rather than the highly specialised tax domain. Building on GerDaLIR, Bönisch and Mehler [23] proposed an SVR-ensemble approach over Transformer-based embedding spaces and observed that the embedding space topology is heavily influenced by passage length, with texts clustering together based on length regardless of semantic content. However, their focus remains on the downstream retrieval methodology, the ensemble classification layer, rather than

³While a vast body of literature focuses on architectural RAG optimisations, such as query expansion, hybrid search methodologies, or neural re-ranking, these techniques often act as compensatory layers that mask the representational deficiencies of the base embeddings [27]. Therefore, investigating the root cause within the embedding space itself remains a critical, yet under-explored, necessity.

on evaluating modern embedding models or diagnosing and improving the intrinsic quality of the embedding space itself.

3.3 Geometric Properties of Embedding Spaces

While the theoretical foundations of representation space geometry have been extensively formalised, prior empirical investigations into these properties have predominantly been confined to general-language domains. Foundational studies by Ethayarajh [21] and Mu and Viswanath [60] exposed that contextualised embeddings from standard language models exhibit severe anisotropy, occupying a narrow cone rather than utilising the full representational capacity of the vector space. Building on these diagnostics, Wang and Isola [22] established alignment and uniformity as measurable quality metrics, proving their predictive value for downstream performance on standard Semantic Textual Similarity tasks.

To address these geometric pathologies, several post-hoc correction methods have emerged in the recent literature. While Li et al. [56] introduced complex normalising flows, approaches such as the ABTT transform [60] and Whitening transformations demonstrated that linear-algebraic interventions can achieve representational improvements [95]. However, these optimisation strategies were almost exclusively validated on general-domain sentence embeddings, leaving their efficacy in highly specialised, dense retrieval settings largely unexplored.

3.4 Research Gap and Contribution

From this review of related work, two major research gaps emerge, which define the contribution of this thesis.

First, while domain-adaptive pre-training is a well-established paradigm for English legal NLP [20, 104], the German legal domain remains under-explored in this specific context. Foundational work has demonstrated the utility of static word embeddings for query expansion [107] and established general LIR benchmarks [23, 24]. For German tax law specifically, recent advancements have focused almost exclusively on the generative component of legal AI systems [5, 78]. Consequently, no prior work has systematically evaluated the suitability of modern, contextualised Transformer-based embedding models for the retrieval stage.

Second, while the intrinsic geometric properties of embedding spaces are well characterised for general-domain models [21, 22] and post-hoc corrections have been shown to improve representational quality [56, 95], their application to specialised legal domains is entirely absent. The question of whether these geometric metrics carry predictive information about downstream retrieval performance in complex domains and can thus serve as lightweight diagnostic proxies in settings where extrinsic human evaluation is costly, remains open.

This thesis addresses both gaps. It provides an empirical characterisation of how general-purpose and domain-adapted embedding models perform specifically in the German tax law domain, combining intrinsic geometric analysis with task-based extrinsic retrieval evaluation within an operational RAG framework. Furthermore, it systematically compares various optimisation strategies, including post-hoc geometric transformations and domain-specific fine-tuning, to investigate the extent to which intrinsic geometric metrics can serve as reliable indicators for retrieval quality. In doing so, it complements the generator-focused work of Wind et al. [5] and Luketina et al. [78] by providing the retriever-focused perspective that has been absent from the German tax law AI landscape.

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Appendix

A Supplementary Corpus Analysis

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B Implementation Details

Put here more Appendix content, e.g. implementation details, code snippets, etc.

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